# HOW SQUALIFY WORKS

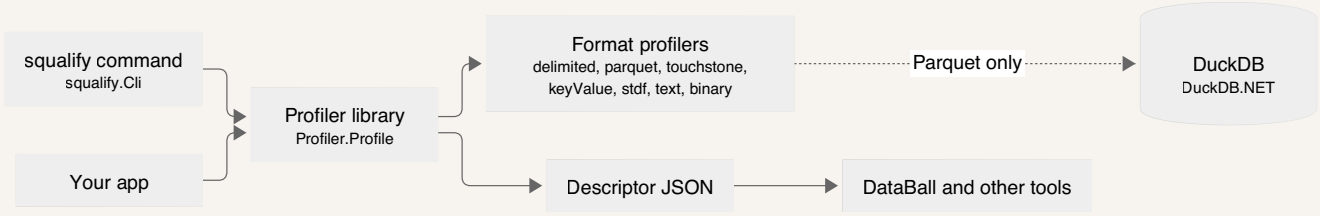
> **Design stage.** This page describes the agreed design (spec pass, 2026-10-13, amended after the sample-data review the same day). Three slices have landed. sq-01 added the CLI shell, `ToolInfo` and `SqualifyException`. sq-02 added the descriptor model, the canonical JSON writer and the determinism kit (`DetMath`, `XxHash64`, `Ordinal`) and the golden-file test harness. sq-03 added the byte pass, the format registry and the generic text and binary profiles, so `squalify profile` now works on any file. Every source file that does not exist yet is marked *(planned)*. When an architectural review or a slice changes how squalify works, this page and its diagrams are updated in the same change (see “Docs” in `AGENTS.md`).

squalify reads one data file and writes a **descriptor**: a JSON summary of what the file is and what is inside it. Other tools, such as DataBall, read the descriptor to decide how to handle the file. squalify never changes the file and never needs the network.

Each diagram shows one idea. Labels are in plain language, with the code name in small text. Source paths are relative to `src/squalify/`, so the CLI project is written `../squalify.Cli/`.

## 1. THE PIECES

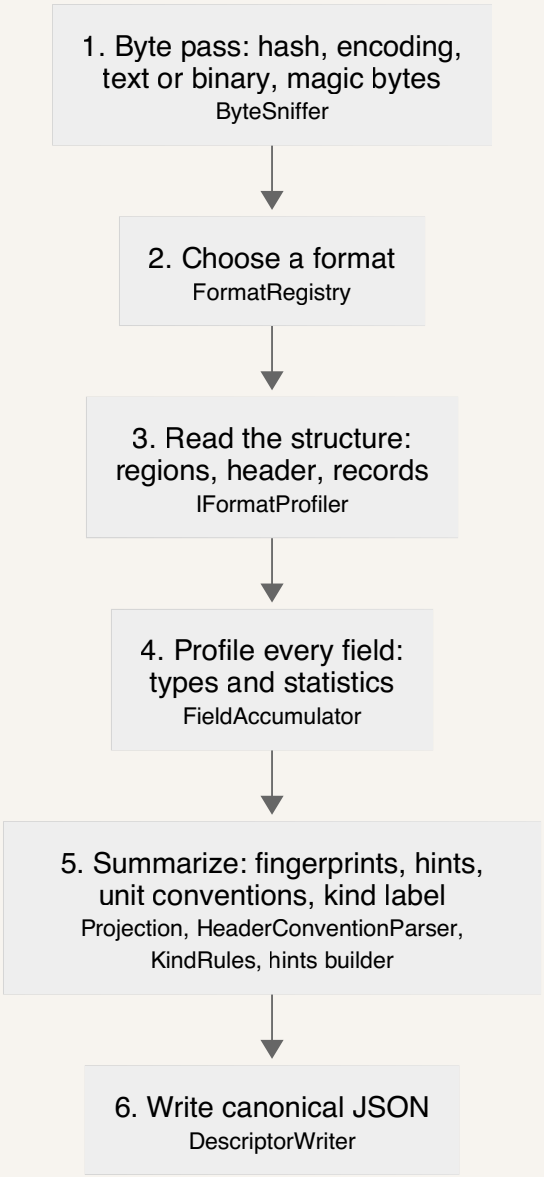
The `squalify` command is a thin shell around the library. Each kind of file has its own profiler. DuckDB is used only to read Parquet. The descriptor is the only thing that leaves the tool.




*Source:* `../squalify.Cli/CommandFactory.cs`, `../squalify.Cli/CliApp.cs`, `ToolInfo.cs`, `SqualifyException.cs`, `Descriptor/`, `Json/DescriptorWriter.cs`, `Profiler.cs`, `Formats/FormatRegistry.cs`, `Formats/GenericTextProfiler.cs`, `Formats/GenericBinaryProfiler.cs`; *the other profilers under* `Formats/` *(planned).* The CLI works today: `squalify --version` prints the tool version, and `squalify profile <file>` writes the descriptor to stdout, or to the `-o` file. Only the `text` and `binary` profilers exist so far, so every file is described as text, binary or empty; the other profilers in the diagram are planned. A directory is an error.

## 2. ONE PROFILE RUN

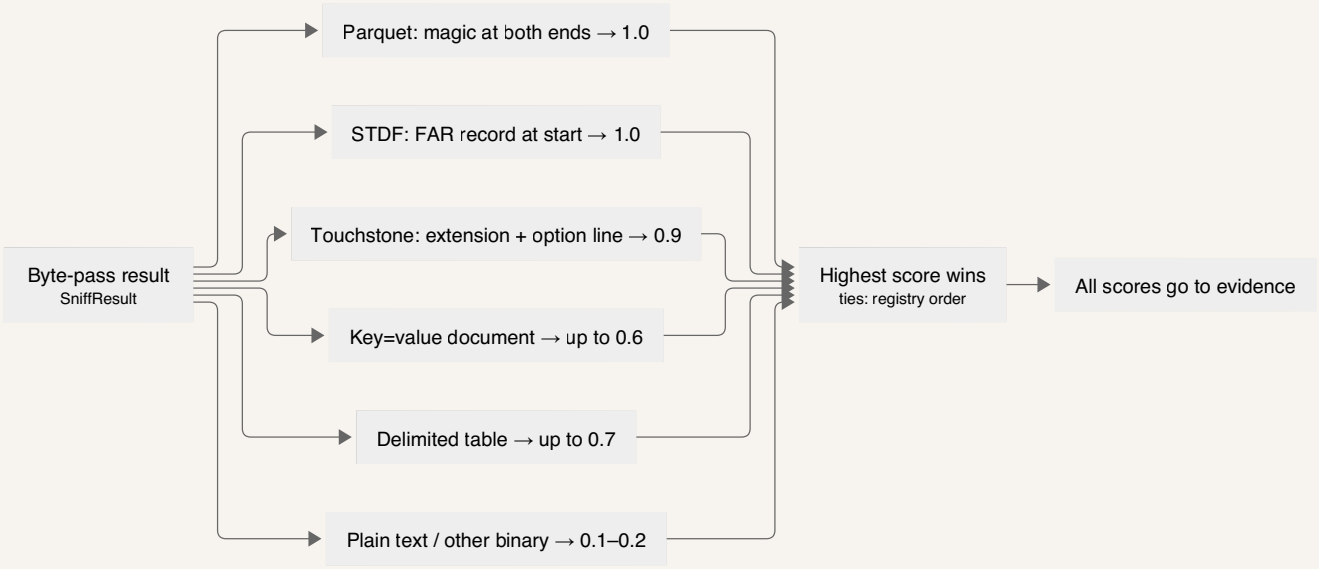
Every file goes through the same steps. A step that reads the file reads all of it, start to end, and keeps only bounded summaries in memory. The file can be far larger than RAM.




*Source:* `Json/DescriptorWriter.cs`, `Profiler.cs`, `Sniff/ByteSniffer.cs`, `Formats/FormatRegistry.cs`, `Formats/IFormatProfiler.cs`; `Fields/`, `Fingerprint/`, `Units/`, `Kind/` *(planned).* Steps 3 to 5 are only partly built: the generic text and binary profiles are the whole of step 3 for now, and the fingerprint and kind label hold fixed placeholder values until their slices land.

## 3. CHOOSING A FORMAT

Every profiler looks at the byte-pass result and gives a score from 0 to 1. The highest score wins, and a tie goes to whichever profiler comes first in the registry list. Every score and its reason are kept in the descriptor's evidence, so you can see why a format was chosen.



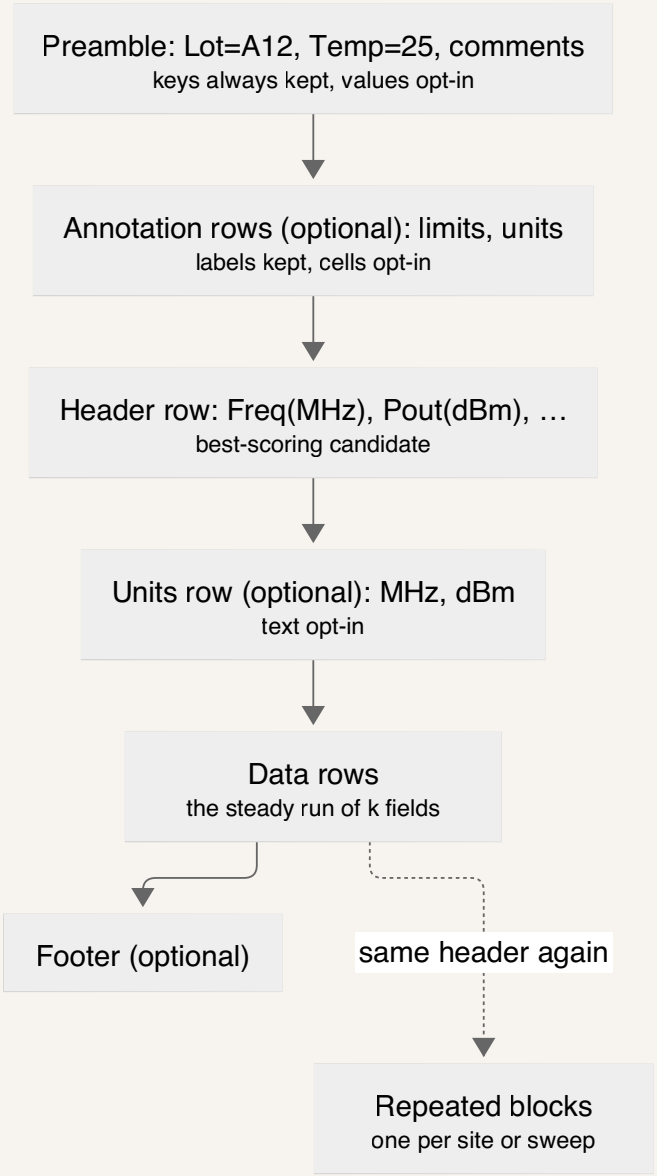

*Source:* `Formats/FormatRegistry.cs` *; the profilers other than text and binary are planned.* A file with no bytes is called empty before any profiler scores it. For files that are not empty, the evidence lists the score of each profiler that is registered, in registry order; today that is just text and binary.

## 4. READING A DELIMITED FILE

A lab CSV is often more than a header and rows. squalify splits the file into regions. It tries every common delimiter at the same time and picks the one that gives the longest steady run of rows with the same field count, and whose cells split cleanly. On tables padded with spaces, splitting on spaces looks almost as steady, so cleanliness decides.

The header is not always the first row of that run. Some exports put rows of limits, units or line numbers above the header, lined up with its columns. squalify scores the first rows of the run and picks the one that

looks most like names over typed data. The rows above it become **annotation** rows: their labels are kept, their cells are opt-in.



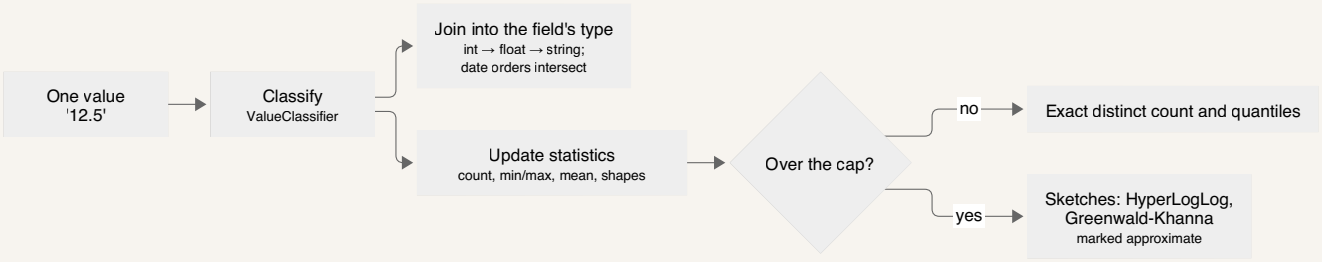

*Source:* `Delimited/DialectSelector.cs`, `Delimited/HeaderDetector.cs`, `Delimited/BlockDetector.cs` *(planned).*

squalify also reports how the table is padded (fixed line length or not) and whether numbers use a decimal comma. It describes a fixed-width table but still reads every row through the tokenizer.

## 5. PROFILING ONE FIELD

Each value is classified on its own: empty, true/false, whole number, decimal number, date, time or text. A field's type is the narrowest type that fits **every** non-empty value. Empty markers such as `NA` or `NA.` come from a fixed, versioned list. Statistics are exact up to a cap. Past the cap they switch to a fixed approximation that gives the same answer every time, and the descriptor marks them as approximate.

Dates never assume a locale. `6/5/2026` could be June 5 or May 6, so squalify reports both possible orders and marks the field ambiguous. It then leaves out the date range and whether the dates rise or fall, because both would assume one order.




*Source:* `Values/ValueClassifier.cs`, `Fields/FieldAccumulator.cs`, `Sketches/` *(planned).*

## 6. WHAT IS ALWAYS INCLUDED, AND WHAT NEEDS `--include-values`

By default a descriptor describes the **shape** of the data and its statistics, not its contents. Literal text from the file appears only when you ask for it: all of it with `--include-values`, or only named keys and columns with `--include-values-for <name>`.

A column with one value for the whole file, such as a serial number, is a special case. Its range would just be that value, so its statistics are opt-in too.

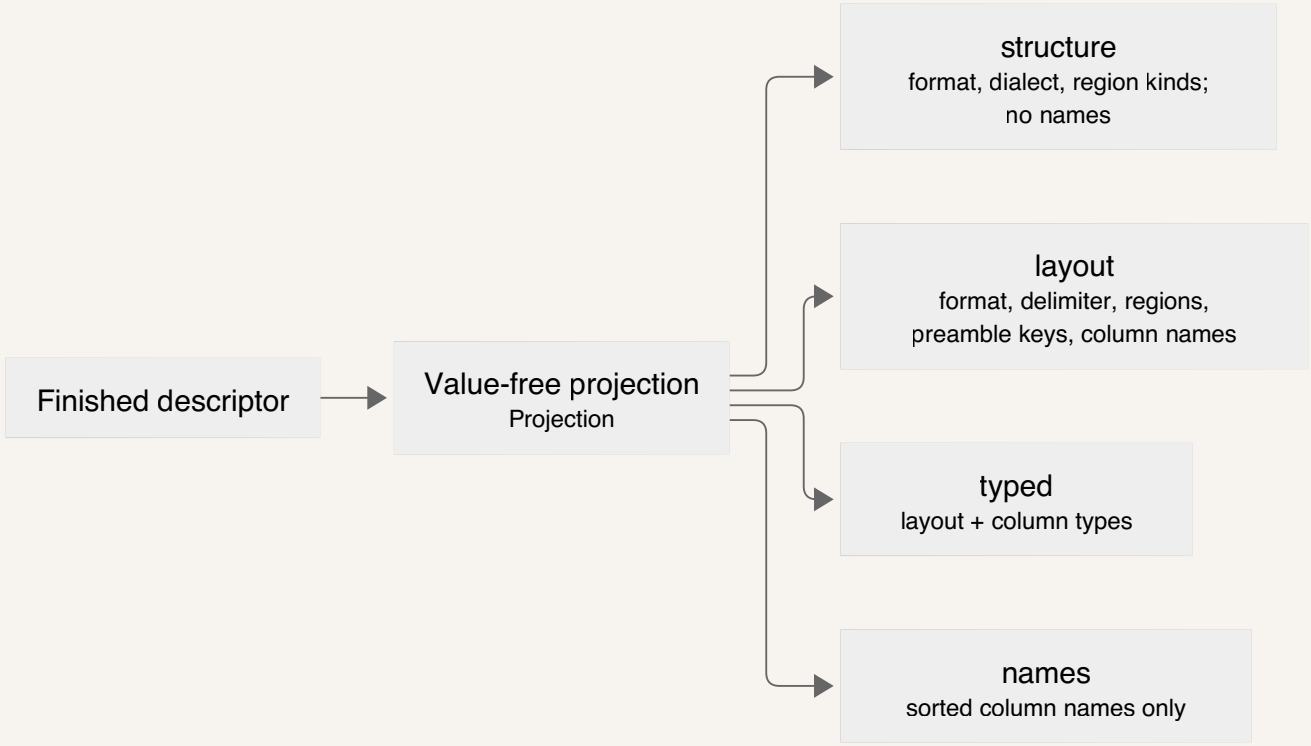
| Always included | Only with --include-values |
|---|---|
| Header names, preamble keys, STDF test names | Preamble and key=value values |
| Types, counts, shapes | Most common values, text min/max |
| Number and date ranges, means, quantiles<br>not for one-value columns | Units-row and annotation cells, comment text |
| Units parsed from header names | Statistics of one-value columns |

*Source:* `ValuesPolicy.cs` *(planned).*

## 7. FINGERPRINTS

squalify hashes a stripped-down copy of the descriptor that contains no data values. Two files of the same type therefore get the same fingerprint, even when their data differs. There are four fingerprints at different levels of strictness. `structure` uses no column names at all, so it survives a changed column set.

A matching fingerprint does not prove two files are the same kind of file. Many unrelated CSVs share one `structure`, and files whose columns change per run get a new `layout` each time. Consumers combine fingerprints with names and hints.

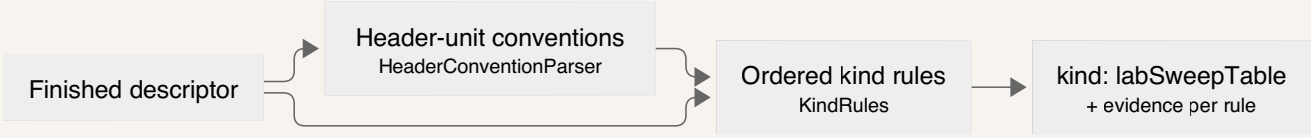



*Source:* `Fingerprint/Projection.cs` *(planned).*

## 8. KIND LABEL AND UNIT CONVENTIONS

After profiling, two small rule sets read the finished descriptor. They never read the file.

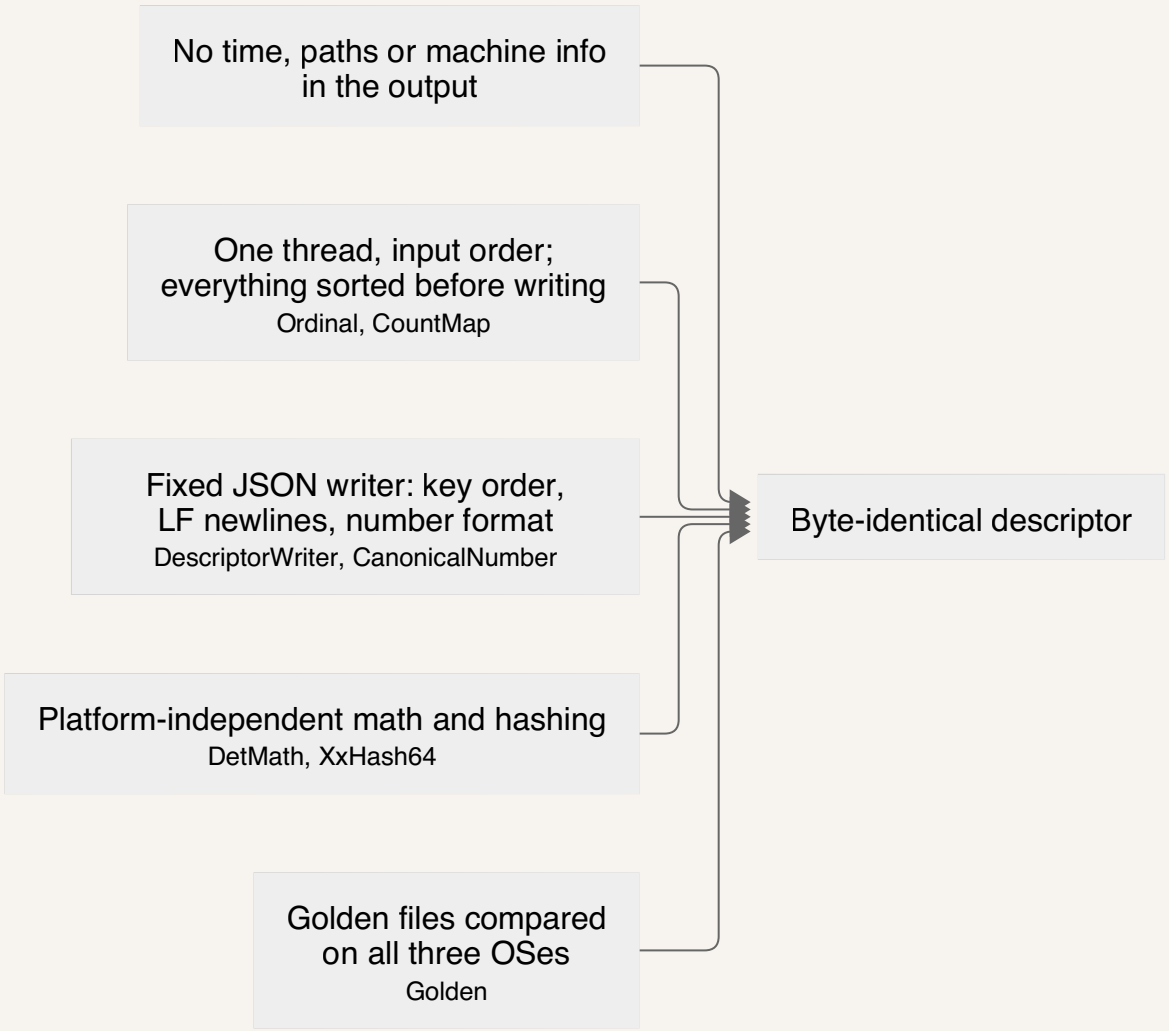
- **Header-unit conventions.** Every column name is checked for how it carries a unit: `I_Total(A)`, `Pout[dBm]`, `VCC_V`, a units row, or none. The most common convention is reported.
- **Kind label.** A fixed, ordered list of rules suggests a generic label, such as `labSweepTable`, `stdfTestLog` or `touchstoneNetwork`. The first rule that matches wins, and every rule's result is kept as evidence.



*Source:* `Units/`, `Kind/` *(planned).*

## 9. WHY THE OUTPUT NEVER CHANGES

The same file and the same squalify version always give the same descriptor, byte for byte, on Windows, macOS and Linux. Five guardrails make that true.

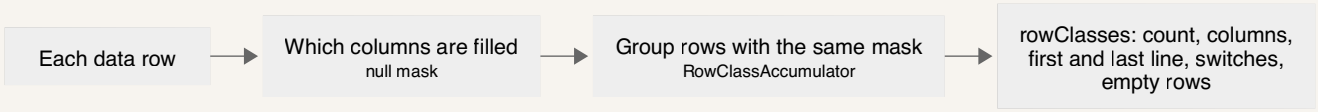



*Source:* `Determinism/Ordinal.cs`, `Determinism/DetMath.cs`, `Determinism/XxHash64.cs`, `Descriptor/CountMap.cs`, `Json/DescriptorWriter.cs`, `Json/CanonicalNumber.cs`; `../../tests/squalify.Tests/Golden.cs`.

Logarithms are the one place where platforms can disagree: the system math library may differ by one bit between operating systems. squalify therefore computes them itself, from additions, multiplications and divisions only, which every CPU rounds identically. A reference table made with an arbitrary-precision tool pins the result bit for bit. Hashing uses squalify's own copy of xxHash64, checked against published test vectors.

## 10. ROW CLASSES

Some exports mix several kinds of row in one table: measurement rows, setup rows, error rows and empty separator rows, switching kind on almost every line. squalify groups rows by which columns are filled in and reports each group, how often the kind switches, and how many rows are empty. Column null counts are also given without the empty rows, so separators do not look like missing measurements.

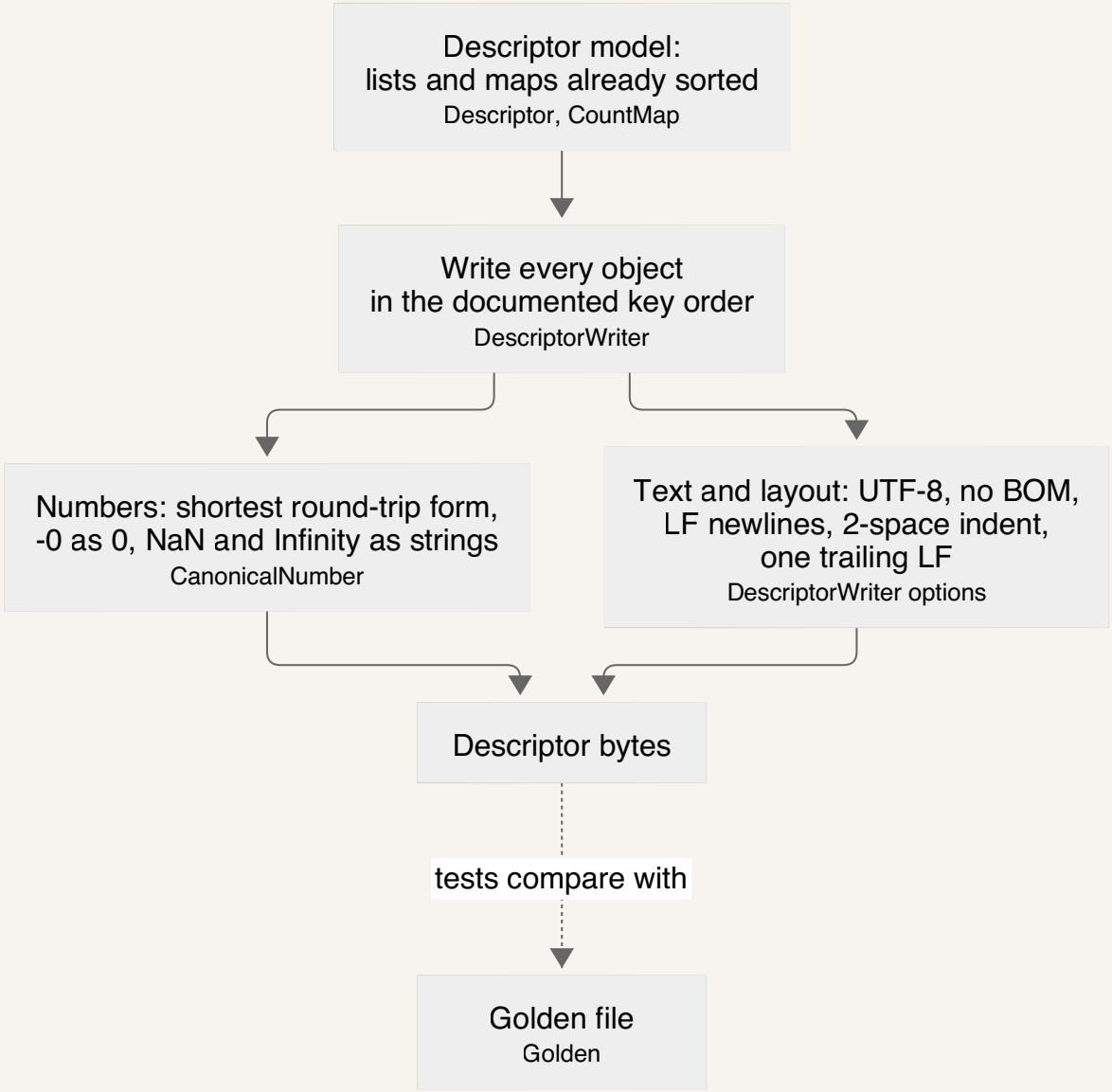



*Source:* `Fields/RowClassAccumulator.cs` *(planned).*

## 11. WRITING THE DESCRIPTOR

The finished descriptor is a tree of plain model objects. The writer turns it into bytes in one fixed way: it writes each object’s keys in the order the schema document lists, and it never sorts, because the model

already holds every list and map in order. Numbers and text go through fixed rules, so the same model gives the same bytes on every machine.

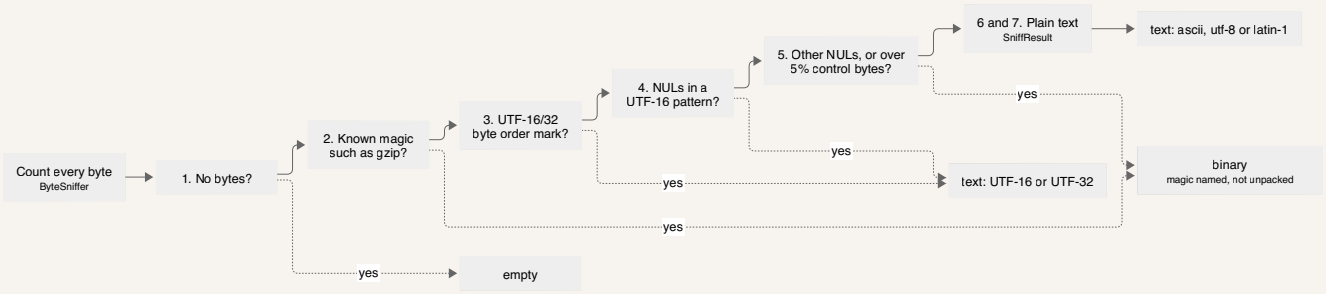



*Source:* `Descriptor/`, `Json/DescriptorWriter.cs`, `Json/CanonicalNumber.cs`; `../../tests/squalify.Tests/Golden.cs`. The key order is part of the schema: see descriptor.md.

## 12. TEXT, BINARY OR EMPTY?

The byte pass counts every byte of the file once and then applies seven rules, in order. The first rule that fits decides. A file is never rejected: odd bytes are counted, and invalid UTF-8 in ascii or utf-8 text is also reported as an issue. A few stray invalid bytes in an otherwise plain text file do not change its encoding,

because a single bad byte in a large log should not turn it into “Latin-1” or “binary”; the allowance is at most 8 per file and at most 1 per 64 KiB.




*Source:* `Sniff/ByteSniffer.cs`, `Sniff/SniffResult.cs`, `Sniff/MagicTable.cs`. The exact thresholds and the magic table are in descriptor.md.